

Math Nuggets for GEN AI

Multi-Head Attention: Capturing Multiple Perspectives Simultaneously

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Beyond Single-Head Attention: The Multi-Perspective Challenge

While self-attention revolutionized sequence modeling, researchers discovered a fundamental limitation: a single attention mechanism could only capture one type of relationship at a time. Consider the sentence:

"The programmer who debugged the complex algorithm efficiently solved the problem."

Multiple relationships exist simultaneously:

- **Subject-verb agreement:** "programmer" $\leftrightarrow$ "solved"
- **Relative clause structure:** "who" $\leftrightarrow$ "programmer"
- **Object relationships:** "algorithm" $\leftrightarrow$ "debugged"
- **Adverbial modification:** "efficiently" $\leftrightarrow$ "solved"

Single-head attention must choose which relationship to prioritize, potentially losing crucial information about other linguistic patterns.

Multi-Head Attention: Parallel Processing of Multiple Relationships

Multi-head attention solves this limitation by running multiple attention mechanisms in parallel, each learning to focus on different types of relationships within the same sequence.

Core Concept: Instead of computing attention once, we compute it h times with different learned projections, allowing the model to attend to information from different representation subspaces simultaneously.

Mathematical Foundation:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (1)$$

$$\text{where head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (2)$$

Each head has its own learned parameter matrices:

- $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$ (Query projection for head i)
- $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$ (Key projection for head i)
- $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ (Value projection for head i)
- $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$ (Output projection)

Multi-Head Attention Architecture

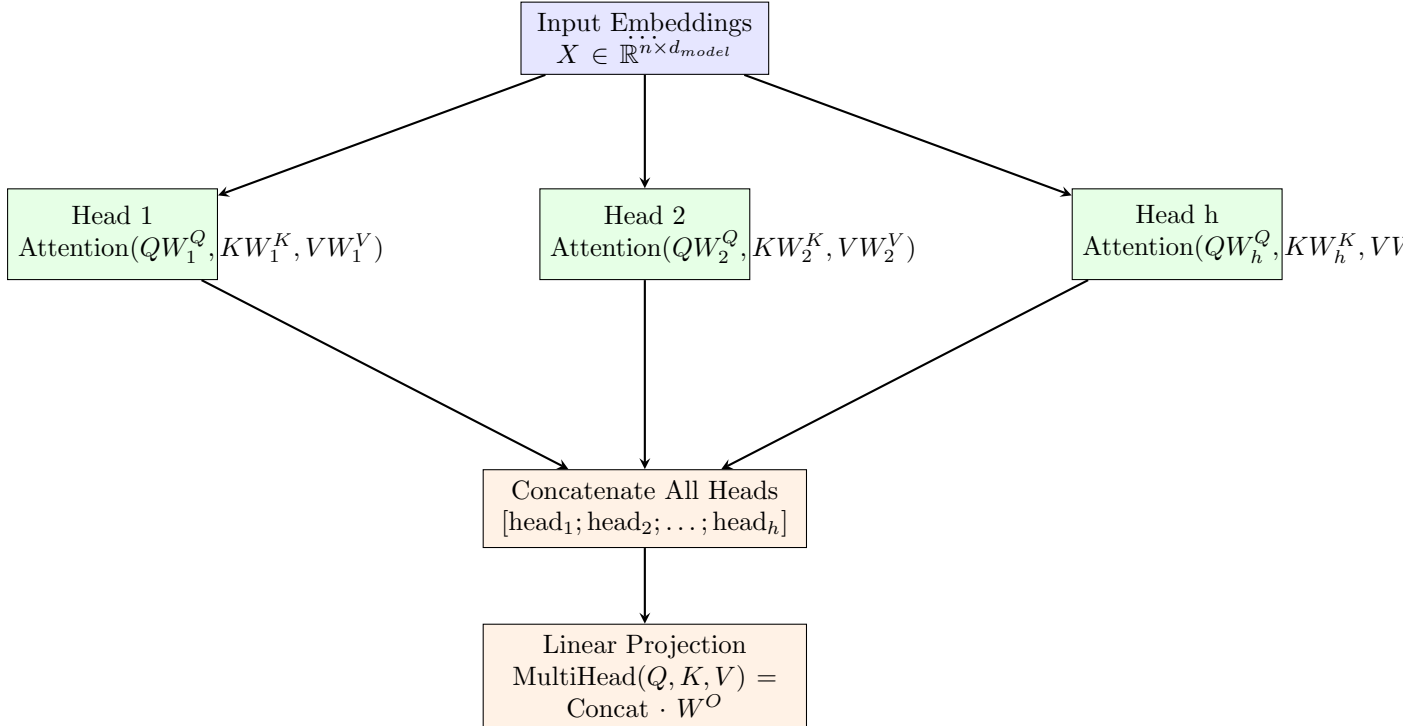


Figure 1: Multi-head attention architecture showing parallel processing of attention heads

Worked Example: 2-Head Attention

Let's compute multi-head attention for the sequence "Alice loves Bob" with 2 heads.

Setup:

- Input dimension: $d_{model} = 4$
- Number of heads: $h = 2$
- Head dimension: $d_k = d_v = d_{model}/h = 2$

Input embeddings:

$$X = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix} \quad (\text{Alice, loves, Bob}) \quad (3)$$

Step 1: Define projection matrices for each head

Head 1 projections:

$$W_1^Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad W_1^K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad W_1^V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (4)$$

Head 2 projections:

$$W_2^Q = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad W_2^K = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad W_2^V = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (5)$$

Step 2: Compute Q, K, V for each head

Head 1:

$$Q_1 = XW_1^Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad K_1 = XW_1^K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad V_1 = XW_1^V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \quad (6)$$

Head 2:

$$Q_2 = XW_2^Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad K_2 = XW_2^K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad V_2 = XW_2^V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad (7)$$

Step 3: Compute attention for each head

Head 1 attention scores:

$$Q_1 K_1^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \quad (8)$$

Head 1 attention weights (after softmax):

$$A_1 = \text{softmax}(Q_1 K_1^T / \sqrt{2}) = \begin{bmatrix} 0.42 & 0.23 & 0.35 \\ 0.27 & 0.37 & 0.36 \\ 0.31 & 0.31 & 0.38 \end{bmatrix} \quad (9)$$

Head 1 output:

$$\text{head}_1 = A_1 V_1 = \begin{bmatrix} 0.77 & 0.58 \\ 0.63 & 0.73 \\ 0.69 & 0.69 \end{bmatrix} \quad (10)$$

Head 2 attention scores:

$$Q_2 K_2^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (11)$$

Head 2 attention weights (after softmax):

$$A_2 = \text{softmax}(Q_2 K_2^T / \sqrt{2}) = \begin{bmatrix} 0.73 & 0.27 & 0.27 \\ 0.27 & 0.73 & 0.27 \\ 0.33 & 0.33 & 0.33 \end{bmatrix} \quad (12)$$

Head 2 output:

$$\text{head}_2 = A_2 V_2 = \begin{bmatrix} 0.73 & 0.00 \\ 0.27 & 0.73 \\ 0.33 & 0.33 \end{bmatrix} \quad (13)$$

Step 4: Concatenate and project

Concatenated output:

$$\text{Concat} = [\text{head}_1; \text{head}_2] = \begin{bmatrix} 0.77 & 0.58 & 0.73 & 0.00 \\ 0.63 & 0.73 & 0.27 & 0.73 \\ 0.69 & 0.69 & 0.33 & 0.33 \end{bmatrix} \quad (14)$$

Final output projection:

$$\text{MultiHead}(Q, K, V) = \text{Concat} \cdot W^O \quad (15)$$

Where $W^O \in \mathbb{R}^{4 \times 4}$ combines information from all heads.

Why Multi-Head Attention Outperforms Single-Head Attention

1. **Multiple Representation Subspaces:** Each head learns different types of relationships
 - Head 1 might focus on syntactic relationships (subject-verb)
 - Head 2 might focus on semantic relationships (word meanings)
 - Head 3 might focus on positional relationships (word order)
2. **Reduced Information Bottleneck:** Instead of forcing all relationships through a single attention mechanism, information flows through multiple parallel channels
3. **Ensemble Effect:** Multiple heads act like an ensemble of attention mechanisms, improving robustness and reducing overfitting
4. **Specialized Learning:** Each head can specialize in different linguistic phenomena without interference from others
5. **Computational Efficiency:** By using smaller dimensions per head ($d_k = d_{model}/h$), total computational cost remains similar to single-head attention while providing richer representations

Empirical Evidence: Research shows that different heads in transformer models naturally learn to focus on:

- Syntactic relationships (subject-object, modifier-head)
- Semantic similarity (synonyms, related concepts)
- Positional patterns (neighboring words, sentence boundaries)
- Coreference resolution (pronouns to their antecedents)

Key Mathematical Insights

Dimension Scaling: The key insight is that $d_k = d_{model}/h$ ensures:

$$\text{Total parameters} = h \times (3 \times d_{model} \times d_k) = 3 \times d_{model}^2 \quad (16)$$

This keeps computational complexity constant while dramatically increasing representational capacity.

Information Integration: The final projection W^O learns to combine insights from all heads:

$$\text{Output}_i = \sum_{j=1}^h W_{i,j}^O \times \text{head}_j \quad (17)$$

This allows the model to weight different types of attention based on context.

Summary

Multi-head attention extends self-attention by enabling parallel processing of multiple relationship types within sequences. This architectural innovation:

- Captures diverse linguistic relationships simultaneously
- Maintains computational efficiency through dimension scaling
- Provides interpretable attention patterns across different heads
- Forms the foundation of transformer architecture success

The mathematical elegance lies in its simplicity: run multiple attention mechanisms in parallel, each learning different aspects of sequence relationships, then combine their insights through learned projections. This approach revolutionized how neural networks process sequential data and enabled the breakthrough performance of modern language models.